Bahria University Lahore Campus

Department of Computer Science

Course Code/number	**GSC-104**
Course Title/Name	**Fundamentals of Mathematics II**
Credit Hours/Contact hours	3
Degree Program	Bachelors of Science in Computer Sciences
Prerequisites or Co-requisites	Mathematics I Basic Algebra
Assessment Methods and Weightage	Quizzes 10% Assignments 20% Mid-Term Examination 20% Final Examination 50% **Total** **100%**
Textbook (or **Laboratory Manual** for Laboratory Courses)	Calculus and Analytic Geometry, Mathematics intermediate part 2, Punjab Text Book Board, Lahore, Pakistan.
Reference Material (With Edition, ISBN#)	1. Calculus, 7th Edition 7th Edition by James Stewart 2. Howard Anton, Calculus, 7th edition John Wiley &Sons. 3. Thomas' Calculus (14th Edition) by George B. Thomas Jr. , Maurice D. Weir , Joel R. Hass
Web Resources/ URL (if any)	
Course Aims	
Course Objectives	The main aim of this course is to introduce students with the fundamental concepts of calculus and analytical

	geometry. In addition to understanding of concepts, one of the key goals is to enable students to apply the learned concepts to solve various practical problems. The course will serve to help the Computer Science students to understand and solve the problems involving Mathematical and logical concepts in other courses and problem domains.
Course Outcomes	After the successful completion of this course the students will learn • To describe real value functions of one and more variables. • To understand the meaning of the derivative in terms of a rate of change and local linear approximation. • Apply the concepts of limits and continuity to solve problems in differential calculus • To understand the meaning of the definite integral both as a limit of Riemann sum to solve a variety of problems. • To understand the basic concepts of Analytic Geometry and its applications. • To understand the concepts of vectors and its applications. •
Course Description/Catalogue	This course is an introduction to the study of Mathematics that develops reasoning and problem solving abilities, with an emphasis on proof. Topics include, Functions of several variables, limits and continuity, Logarithmic differentiation, rate of change, , Integration and its applications, definite and indefinite integrals, area under the curve, analytical geometry, equations for plane, lines in space, linear programming and vectors. This course is intended for students capable of and interested in progressing through the concepts of applied mathematics in more depth and at an accelerated rate.

Lecture Plan (16 Weeks)

Week # and Date	Lecture /Hour #	Topic to be covered	Learning outcomes	Reference Text
1	2	Introduction and Functions.	Introduction to the course & Functions	Textbook and Lecture Handouts
	1	Functions (Domain, Range) Function representation and types.	Students will learn concepts of function and its representation.	Textbook and Lecture Handouts
2	2	Composite of functions, Inverse of a function	Students will learn how to compose multiple function into one	Textbook and Lecture Handouts
	1	Function of several variables, , Graph of functions	Students should be able to draw the function graphically.	Textbook and Lecture Handouts
3	2	Limits of function	Students can understand the concept of limit	Textbook and Lecture Handouts

		1	Laws of limit, Indeterminate forms of limits	Student will be able to learn the limit laws.	Textbook and Lecture Handouts
	4	2	Continuity and Discontinuity of functions.	An understanding on the continuity and discontinuity.	Textbook and Lecture Handouts
		1	Continuous and Discontinuous functions and their applications.	A clear understanding about when the function will be continuous and discontinuous with applications.	Textbook and Lecture Handouts
	5	2	Introduction to Differentiation, Rate of change	Grasp the concept of differentiation.	Textbook and Lecture Handouts
		1	Product rule, quotient rule and The chain rule	Understanding rules of differentiation.	
	6	2	Derivatives and its practice problems	Able to use the concept of differentiation to different problems.	Textbook and Lecture Handouts
		1	Implicit Differentiation	Ability to learn and solve implicit differentiation	Textbook and Lecture Handouts
	7	2	Differentiation and its applications	Learn to find out derivatives of logarithmic functions.	Textbook and Lecture Handouts
		1	Differentiation practice problems	Able to use the concept of differentiation to different problems.	Textbook and Lecture Handouts
	8	2	Review and Revision	Reinforcement of above topics	Textbook and Lecture Handouts
		1	Review and Revision	Uninformed assessment to revise all concepts.	Textbook and Lecture Handouts
			Mid-Semester Exam		
	9	2	Derivatives of trigonometric functions	Learn how to apply derivatives on trigonometric functions.	Textbook and Lecture Handouts
		1	Applications and problems of trigonometric functions	Application of differentiation on trigonometric problems.	Textbook and Lecture Handouts
	10	2	Introduction to Integration, definite and Indefinite integrals	Learn the basic concepts of integration, also the definite and indefinite integral.	Textbook and Lecture Handouts
		1	Indefinite Integrals	To learn the techniques to solve indefinite integrals.	Textbook and Lecture Handouts

	11	2	Definite Integrals and Its applications	A good understanding of integral calculus with concept and definite integrals.	Textbook and Lecture Handouts
		1	Definite Integrals and area problems	Understanding and Problems of definite integrals and area problems.	Textbook and Lecture Handouts
	12	2	Introduction to Analytic geometry, distance formula, ratio formula, slope and its applications	Students will learn the analytic geometry and some basic concepts.	Textbook and Lecture Handouts
		1	Types of Equations of line	Ability to use the different types of equations of line.	Textbook and Lecture Handouts
	13	2	Linear Programming , Solution of linear inequalities	Learn how to find solution of linear inequalities by using graph.	Textbook and Lecture Handouts
		1	Linear Inequalities and its applications	Students will learn how to find solution of linear inequalities by using graph.	Textbook and Lecture Handouts
	14	2	Introduction to conic section, circle, tangent line, normal line, parabola.	A brief introduction to Conic section and shapes.	Textbook and Lecture Handouts
		1	Hyperbola, Ellipse,	A brief introduction to hyperbola and ellipse.	Textbook and Lecture Handouts
	15	2	Scalars and vectors	Students will learn the Vector theory and its applications.	Textbook and Lecture Handouts
		1	Vectors and its applications	Students will learn the Vector theory and its applications.	Textbook and Lecture Handouts
	16	2	Review and Revision	Reinforcement of above topics	Textbook and Lecture Handouts
		1	Review and Revision	Uninformed assessment to revise all concepts.	Textbook and Lecture Handouts
			Final Exam		

Semester Calendar for Assignments/Quizzes/Project

Assignments/ Projects and quizzes Plan	Week #	Assignment No.	Quiz No.	Project Title	Assignment/Project Quiz Date	Result Date of Assignment/Project/Quiz
	1					
	2	Assignment-1				
	3		Quiz-1		Assignment-1	
	4					Assignment-1, Quiz-1
	5	Assignment-2				
	6		Quiz-2		Assignment-2	
	7					Assignment-2, Quiz-2
	8					
	9					
	10	Assignment-3	Quiz-3			
	11				Assignment-3	
	12	Assignment-4	Quiz-4			Assignment-3, Quiz-3
	13				Assignment-4	
	14					Assignment-4, Quiz-4
	15					Last week
	16					

Instructor Name: Assad Sajjad

Instructor Signature: Assad Sajjad

Date 21-03-2023